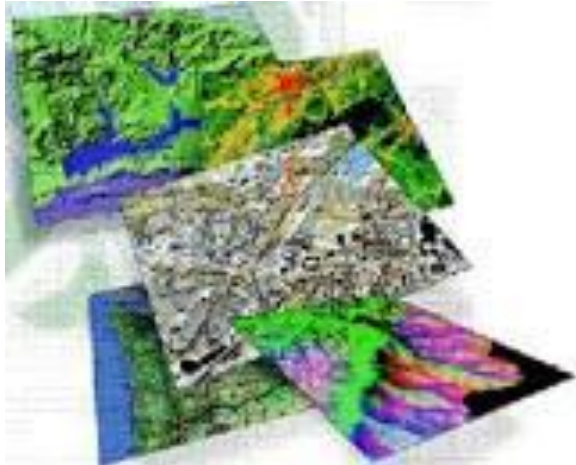


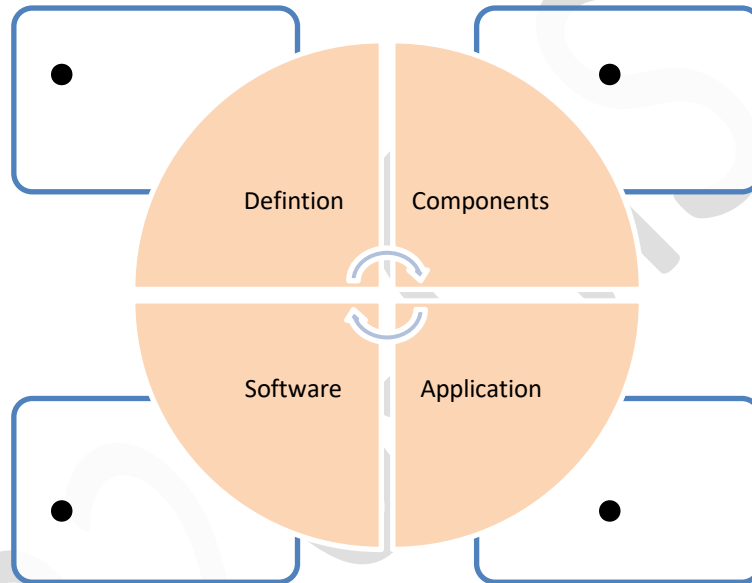
LEARNING MODULE NO. 1

Title	OVERVIEW OF GEOGRAPHIC INFORMATION SYSTEM 
Topic	INTRODUCTION TO GIS: DEFINITIONS, TYPES, COMPONENTS, AND APPLICATIONS
Time Frame	5 hrs.
Introduction	This course introduces the fundamental theories and concepts of Geographic Information Systems (GIS). The course content will include the definition, benefits, applications and six parts of GIS. Geographic Information Systems (GIS) are systems designed to collect, store, analyze, and visualize spatial or geographic data, helping users understand relationships and patterns in a specific area. GIS data is typically categorized into two types: raster (grid-based images like satellite photos) and vector (point, line, and polygon features). The key components of GIS include hardware, software, data, people, and procedures. GIS is widely applied in areas such as urban planning, environmental management, agriculture, and transportation, making it a crucial tool for decision-making and spatial analysis.
Objectives	<i>In this module, learners will be able to:</i> 1. Define Geographic Information Systems (GIS) and explain their significance across various fields by describing diverse real-world applications, outlining essential components, and differentiating between various types of GIS data, including their unique characteristics and uses.

Learning Activities
(to include
Content/Discussion
of the Topic)



Draw a concept map on the definitions, components, applications, and software used in GIS.

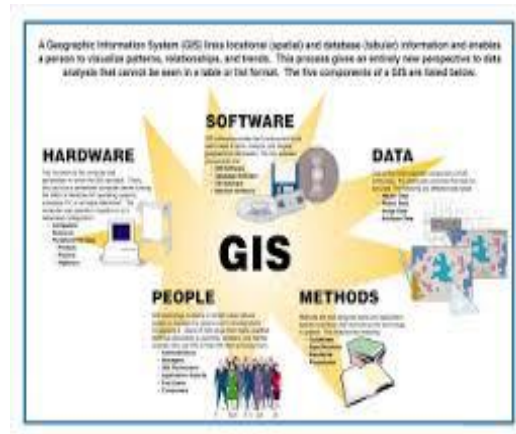


What is GIS?

GIS stands for **Geographic Information System**, is a computerized tool that captures, manipulates, stores, presents, and manages the data related to geography. It was coined by **Roger Tomlinson** in the year 1968 and was given the title "**The Father of GIS**". GIS helps in decision-making, solves problems related to other geographic issues, and visualizes spatial environmental data. It represents the data related to space in form of maps like Google maps, Apple maps, and open street maps. This map includes spatial data in terms of where the physical things are exactly located, their quantities, densities, routes, nearby location, etc. GIS software applications can be downloaded on the user's smartphone or can be viewed on a computer system, which allows the user to perform analysis, spatial information, create searches, store data, edit data, and share the data visually. Few of the GIS applications are GMaps and OMaps.

Definitions:

- A **geographic information system (GIS)** integrates hardware, software, and data for **capturing**, **managing**, **analyzing**, and **displaying** all forms of geographically referenced information.

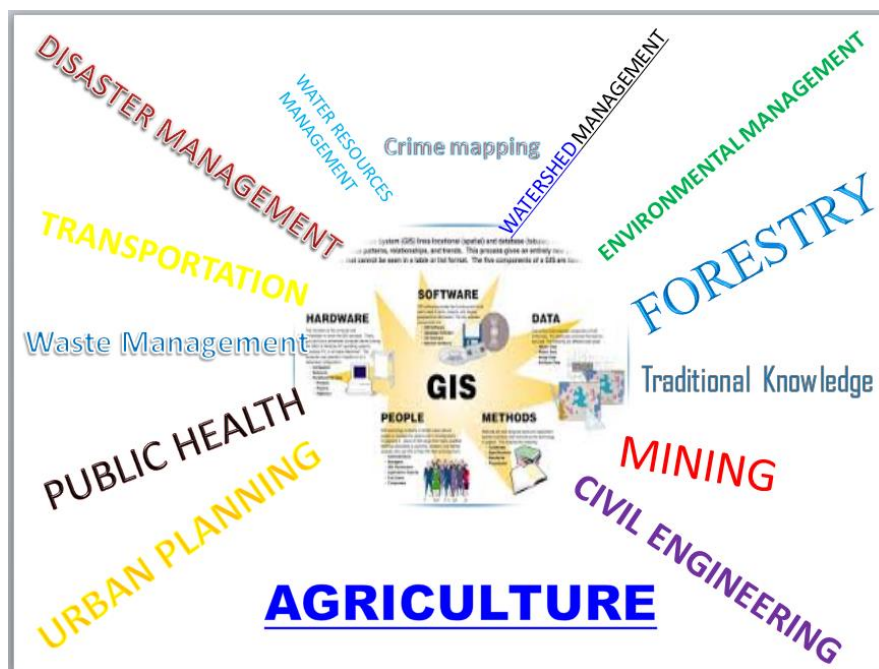


- Advances in **Geographic Information Systems (GIS)** technology and the mathematical/statistical tools for modeling and simulation have led to the growing application of quantitative techniques in many areas of the earth sciences.
- Primary analysis tool in landslide hazard mapping as it is useful for spatial data management and manipulation.
- Excellent alternative to conventional mapping techniques in monitoring and mapping geo-hazards areas, these tools used to investigate, monitor and assess geo-hazards through the building of a geographical database.
- Voluminous geographic data have been, and continue to be collected with modern data acquisition techniques such as global positioning systems (GPS), high-resolution remote sensing, location-aware services and surveys, and internet-based volunteered geographic information.
- The integration of GIS technology in an information system is one of the approaches that could be used to strengthen the communication between the government body and the community to help mitigate the risk in the future.

Top Five Benefits of GIS

- Cost Savings and Increased Efficiency
- Better Decision Making
- Improved Communication
- Better Recordkeeping
- Managing Geographically

Applications of GIS





Use of GIS in Agriculture

- GIS are extensively used in agriculture, especially in precision farming.
- GIS is used in decision making such as what to plant and where to plant using historical data and sampling.
- Farm Management
- Pest/Disease tracking
- Crop Monitoring
- Yield prediction
- Soil Analysis



Uses of GIS Cloud



How can GIS be used in farming?

GIS can be used by **agricultural agencies to support pesticide and food safety regulations**, show economic impacts of policy, reveal environmental health issues, depict animal health and welfare issues, record data about an area, and arbitrate land use conflicts. GIS is an effective, proven technology in government.

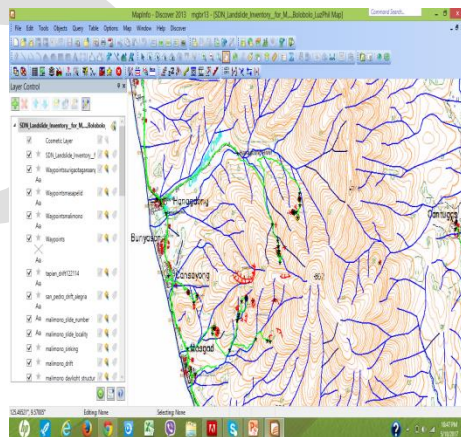


Mining companies use GIS to actively monitor the environmental impacts that may be caused by their activities and conduct reclamation. GIS is **used for monitoring and reclamation** by analyzing and mapping soils, vegetation, surface hydrology, and ground- water.

Mining companies use GIS to

- Target mineral exploration.
- Evaluate mining conditions.
- Model mine construction.
- Display geochemical and hydrology data.
- Improve facility management and policing.
- Apply for mining permits.
- Assess environmental impact.
- Manage land titles.
- Process closures.
- Plan reclamation activities.
- Improve community education.

<https://www.esri.com/content/dam/esrisites/sitecore-archive/Files/Pdfs/library/brochures/pdfs/gis-sols-for-mining.pdf>

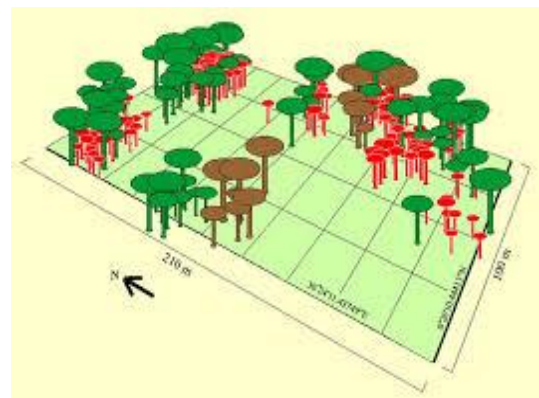
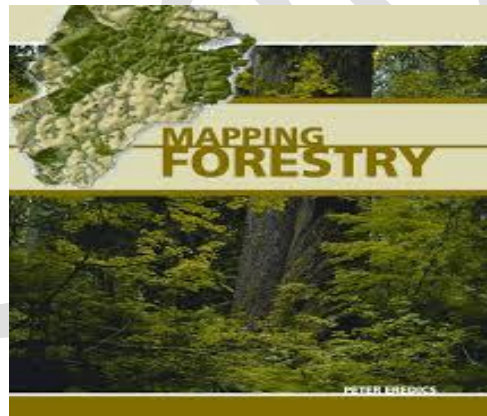


Using GIS in Forestry

GIS enhances forest management by assisting land owners and forest managers to evaluate and analyze the species diversity, age and size of timber, timber density, and volume. We are able to store this data in a geographic information system which allows us to generate forest management plans, prescribe a timber sale, update inventories based on a harvest, and assess for insect and disease threat. (<https://www.foreconinc.com/gis-and-forest-management>)

Remote sensing and GIS are complementary technologies that, when combined, enable improved monitoring, mapping, and management of forest resources (Franklin 2001). The information that supports forest management is stored primarily in the form of **forest inventory databases** within a GIS environment.

GIS and remote sensing are crucial elements of forest management as well. This **technology helps understand the land cover, availability, area, and human encroachment of protected forest areas**. This information plays a very important role in forest management and decision making.



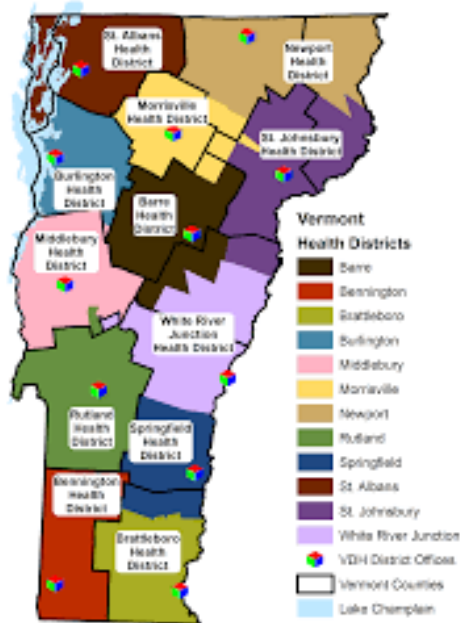
GIS Uses in Public Health

GIS has continued to be used in public health for **epidemiological studies**. By tracking the sources of diseases and the movements of contagions, agencies can respond more effectively to outbreaks of disease by identifying at-risk populations and targeting intervention.



How is GIS used in healthcare?

Public health professionals use GIS to **analyze chronic disease trends** (e.g., heart disease, diabetes, cancer), analyze access to public health services (e.g., vaccinations), analyze the built environment, respond to natural and man-made disasters, and design community health communications programs.



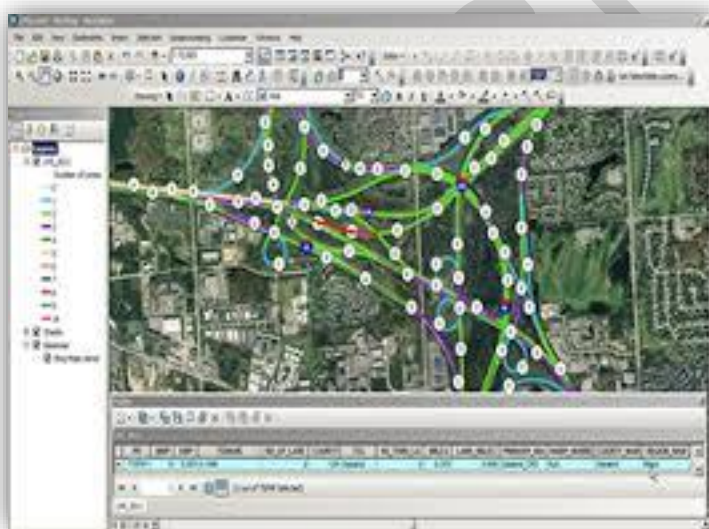
- GIS is a public health tool

Disease Outbreak

Monitoring purposes

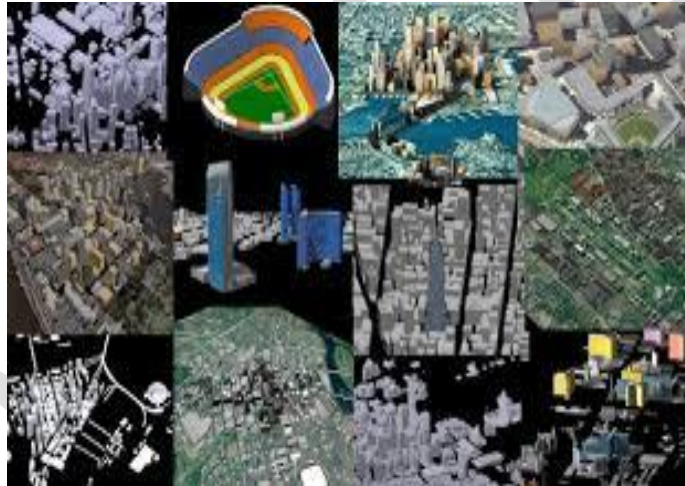
GIS in Transportation

- In transportation profession the world have discovered and embraced GIS as an important tool, which helps user in managing, planning, evaluating and maintaining transportation system.
- A specific branch of GIS called GIS-T, has been used for diverse purposes, which refers to the principles and application of applying geographic information technology to transportation problem.



GIS APPLICATIONS IN CIVIL ENGINEERING

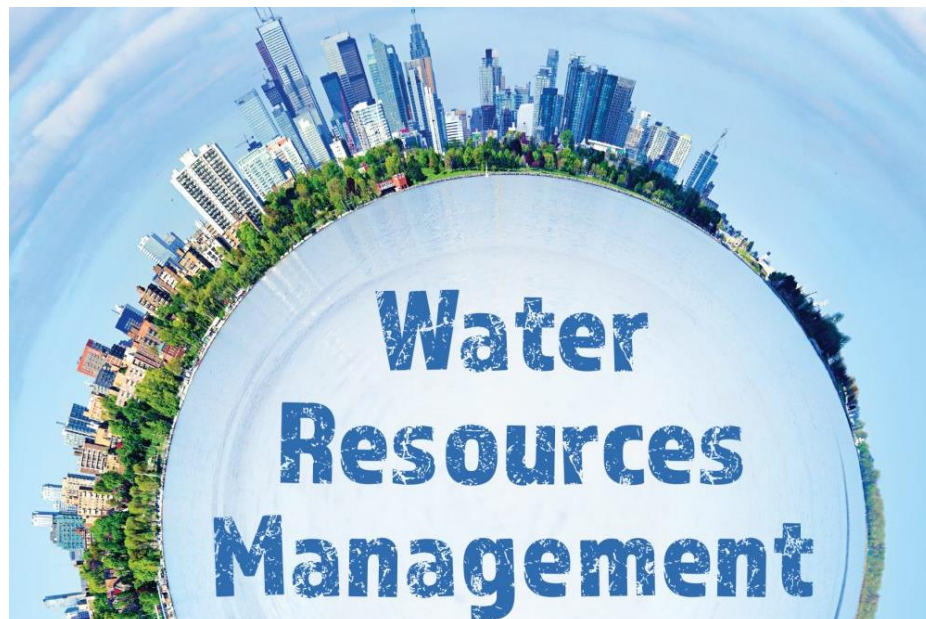
- There are many fields in Civil Engg where GIS is being employed as a tool to analyze, design and implement effective and efficient solutions.
- GIS functionalities have been used to assist in the analysis, selection prioritization and implementation in Civil Eng'g projects.
- Many GIS applications have been developed in Hydrology, Hydraulics, Surveying, Environmental, water resources, Transportation, Geotechnics and other fields in Civil Eng'g.



Using GIS in Watershed Management

- is the study of the relevant characteristics of a [watershed](#) aimed at the sustainable distribution of its resources and the process of creating and implementing plans, programs, and projects to sustain and enhance [watershed](#) functions that affect the [plant](#), [animal](#), and [human](#) communities within the watershed boundary. Features of a watershed that agencies seek to manage include [water supply](#), [water quality](#), [drainage](#), [stormwater](#) runoff, [water rights](#), and the overall planning and utilization of watersheds. [Landowners](#), [land use](#) agencies, stormwater management experts, environmental specialists, water use surveyors and communities all play an integral part in watershed management.





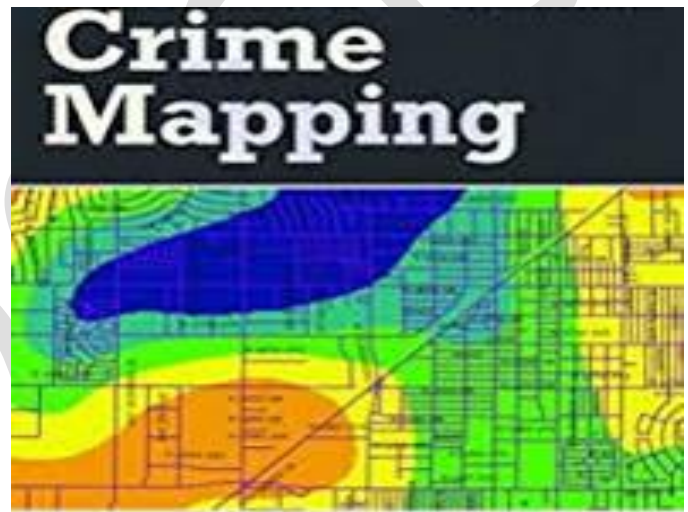
Water resource management is the activity of planning, developing, distributing and managing the optimum use of [water resources](#). It is a sub-set of [water cycle management](#). Ideally, water resource management planning has regard to all the competing demands for [water](#) and seeks to allocate water on an equitable basis to satisfy all uses and demands. As with other [resource management](#), this is rarely possible in practice.



GIS and CRIME MAPPING

Crime mapping is used by analysts in [law enforcement agencies](#) to map, visualize, and analyze crime incident patterns. It is a key component of [crime analysis](#) and the [CompStat](#) policing strategy. Mapping crime, using [Geographic Information Systems](#) (GIS), allows crime analysts to identify [crime hot spots](#), along with other trends and patterns.

Using GIS, you can **create a map that can identify where the crimes are occurring and clarify what crimes are or are not related based on your research**. This can allow investigators to target their efforts and line officers to patrol and respond to locations while being more fully aware.



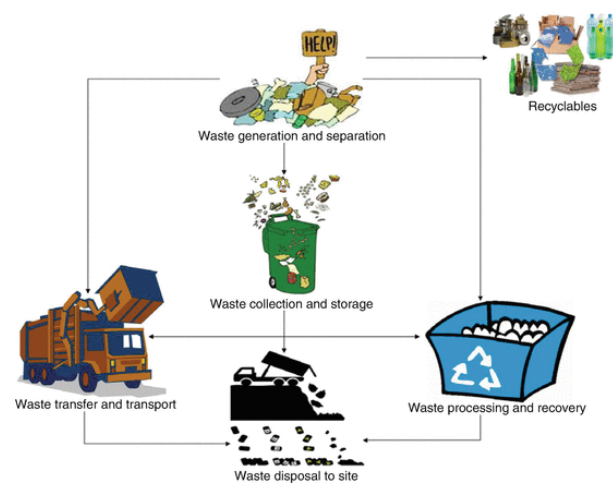


Why GIS can be useful for waste management?

GIS and application of **multi-criteria analysis** can **produce suitable sites for solid wastes disposal** which is a vital component in solid waste management. Global positioning system used for earth observation can track waste mainly by tracking vehicles, or paperwork chains.

How is GIS used in waste management?

GIS can add value to waste management applications by **providing outputs for decision support and analysis** in a wide spectrum of projects such as route planning for waste collection, site selection exercises for transfer stations, landfills or waste collection points.



Traditional Knowledge (TK)

- TK associated with the biological resources is the knowledge about a country's biodiversity; the applied uses and applications of biological resources and the prevalent practices.
- It is an intangible component of the resource itself. TK has the potential of being transformed into commercial opportunity, providing useful leads for development of products and processes.
- Hence, a share of benefits must accrue to creators and holders of Traditional Knowledge (TK).

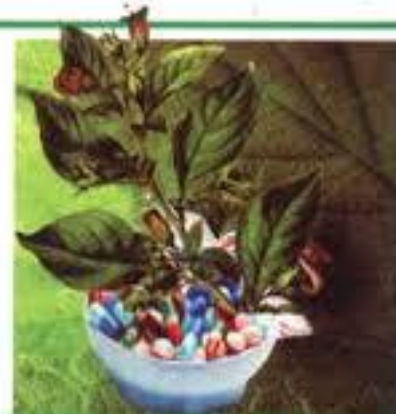
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Traditional knowledge geographic information systems (GIS) are the **data, techniques, and technologies designed to document and utilize local knowledges** in communities around the world. Traditional knowledge is information that encompasses the experiences of a particular culture or society.



Medicinal Plants

Biodiversity, Conservation and Traditional Knowledge



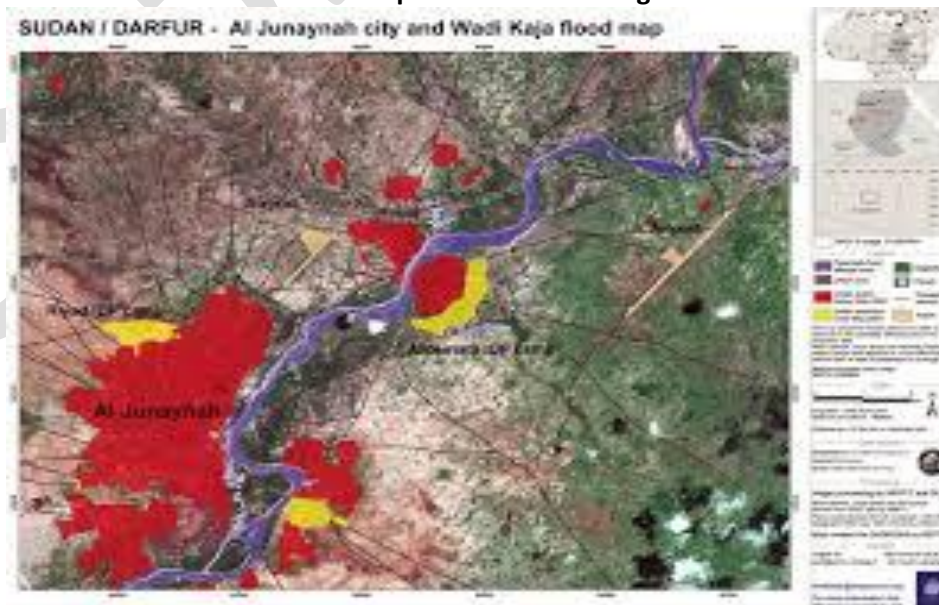
DISASTER MANAGEMENT



- **Disaster**
 - This is an event causing great damage, injury or loss of life.
- **Disaster Management**
 - Can be defined as the discipline and profession of applying science, technology, planning and management to deal with extreme events.
- The emphasis of disaster management is prevention and loss reduction.

❖ Disaster management (ex: landslide, floods, drought)

GIS-based Map for Disaster Management



URBAN PLANNING

- **Urban**
 - An Area with a relatively high population density that contains a set of closely related activities.
- **Planning**
 - A Management Tool in the sense of setting objectives and targets for production.
- **Urban Planning**
 - Decision making for the allocation of Land Uses in a system where the Market is the primary mode.
- Traffic Analysis
- Assessing physical transformation of cities
- Land suitability modeling
- Delineating environmental sensitive area.
- Planning zones

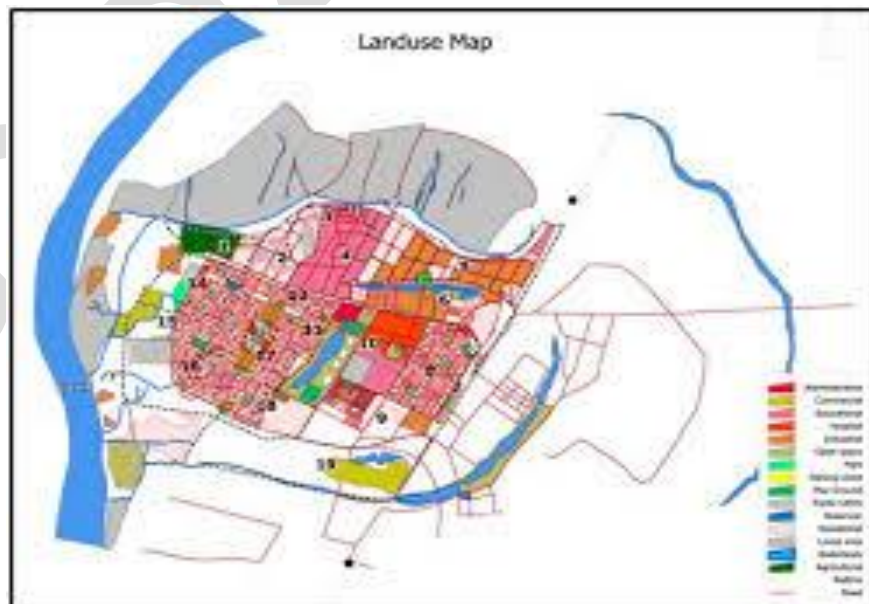


Fig-1

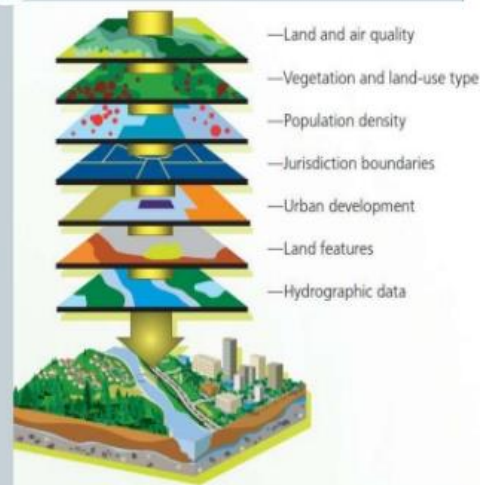


- **Environmental management system (EMS)** refers to the management of an organization's [environmental](#) programs in a comprehensive, systematic, planned and documented manner. It includes the organizational structure, planning and resources for developing, implementing and maintaining policy for [environmental protection](#).

GIS IN ENVIRONMENT STUDIES

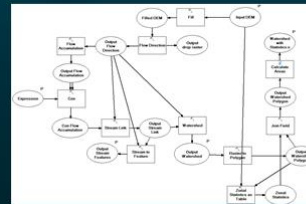
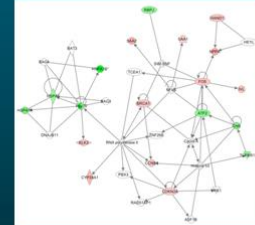
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- GIS is a powerful tool for environmental data analysis and planning.
- A digital basemap can be overlaid with data or other layers of information onto a map in order to view spatial information and relationships.



Six Parts of a GIS

1. Hardware
2. Software
3. Data
4. Methods
5. People
6. Network



Hardware

- Powerful computers
- Printing/Plotting and other visualization capabilities
- Input devices
- Large amounts of disk space



Hardware is perhaps the simplest part of a GIS to explain. The hardware part of GIS refers to powerful computers, printers, output devices, such as monitors, input devices, and large amounts of disk space to store the, typically, large GIS data sets. A typical GIS computer workstation, has a powerful CPU, large amounts of RAM, large monitors, fast network connectivity, and large amounts of disk space.

Software

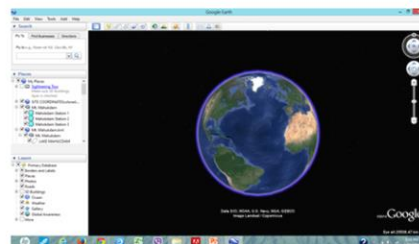
- Provides tools to:
 - Manipulate
 - Store
 - Query
 - Analyze
- There are many free and commercial software packages.



Spatial Information

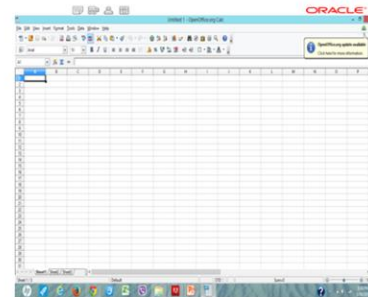
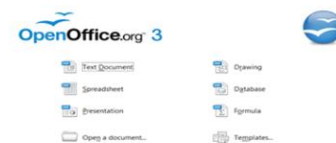
The *software* part of the GIS refers to the many free and commercial software packages that allow a GIS user to perform GIS functions such as the manipulation, storage, query, and analysis of spatial and non-spatial information.

Software Tools

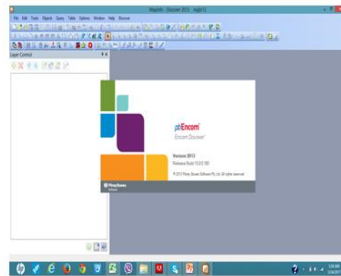


MAPWINDOW

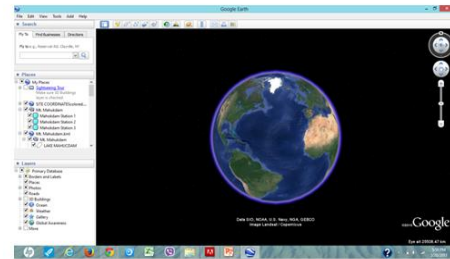
7-Zip



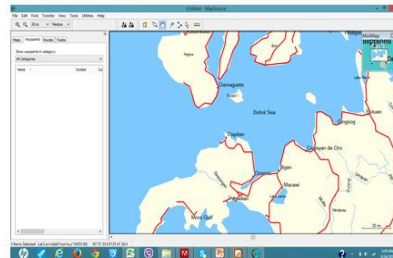
Another Software Tools



MapInfo



GoogleEarth



MAPSOURCE



Data



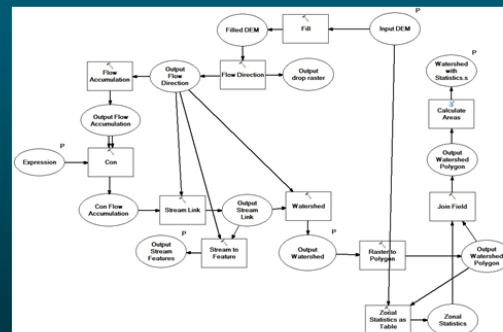
- A GIS can ingest any type of data
 - Both spatial and non-spatial
- We are currently in “Data Overload”
 - Visualizing data helps humans weed through it all
- Half of all time spent on a GIS projects will simply be working with data

The *data* part of a GIS, refers to any and all spatial and non-spatial information stored in a digital format on the computer. A GIS can ingest any type of data in many different types of formats. The ability for a GIS to consume such a wide range of data can often produce the effect of “data overload”. With the wide proliferation of sensors (data collectors), there is more data available to us today than ever before. With so much data, it is often difficult for a person to easily understand what the data actually is (as if its useful). A GIS can visualize the data as a map, graph, or other graphic, that helps humans understand and weed through massive amounts of data.

In fact, the problem of “too much data” does require a GIS user to spend quite a bit of time managing the data. In general, half of all the time you spend on a GIS project, will simply be working with and preparing the data for maintenance, storage, analysis, and output.

Methods

- Formulas
- Statistics
- Analysis
- Algorithms



The methods part of a GIS refers to the formulas, statistics, analysis, and algorithms that are used to turn data into information, so that humans can turn that information into knowledge through interpretation.

People

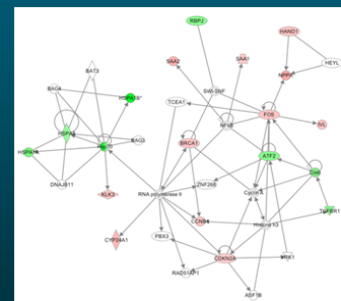
- Skilled GIS Professionals
- Skilled GIS Users
- Novice GIS Users
- .
- .
- .
- Public Users



The *people* segment refers to anyone that uses a GIS. This includes the general public that may be using a printed map, or a digital globe, all the way up to skilled GIS professionals, that build new functionality and push the limits of what is possible in a GIS.

Network

- Dissemination of data
 - Transfer datasets
 - Collaboration
- Display of information
 - Web maps
 - Web Applications



Lastly, *network* refers to both the computer, and social network. Both of these networks assist in the dissemination of data. Where the dissemination of data is through transferring of data sets or collaboration, sharing the data from a GIS is a very common and useful operation. Additionally, these networks allow for the display of information in the form of web maps, web applications, or even paper maps using our social network.

Types of GIS Data Explored: Vector and Raster

A geodatabase is a database that is in some way referenced to locations on the earth. Coupled with this data is usually data known as attribute data. Attribute data generally defined as additional information, which can then be tied to spatial data.

What types of GIS Data are there?

GIS data can be separated into two categories: spatially referenced data which is represented by *vector and raster forms (including imagery)* and attribute tables which is represented in *tabular format*.

Within the spatial referenced data group, the GIS data can be further classified into two different types: **vector** and **raster**.

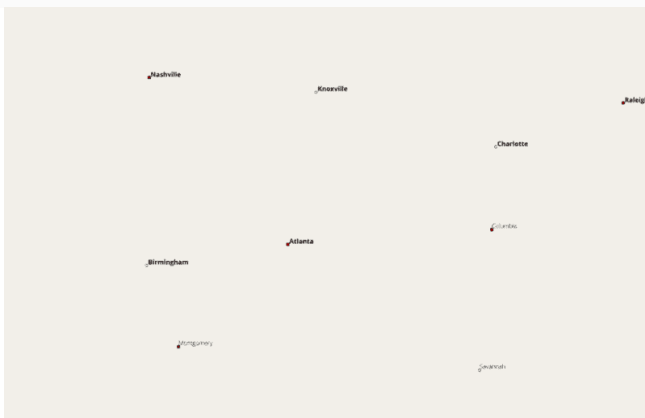
Most GIS software applications mainly focus on the usage and manipulation of vector geodatabases with added components to work with raster-based geodatabases.

VECTOR DATA

Vector data is split into three types: **point, line (or arc),** and **polygon data**.

Point Data

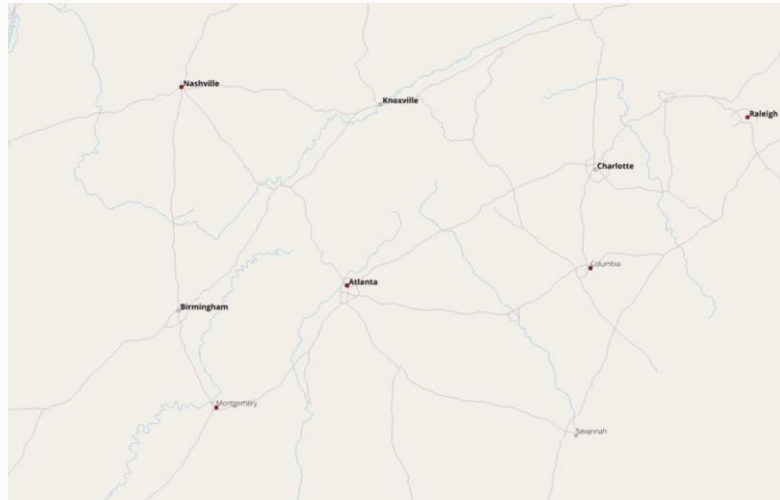
Point data is most commonly used to represent nonadjacent features and to represent discrete data points. Points have zero dimensions, therefore you can measure neither length or area with this dataset. Examples would be schools, points of interest, and in the example below, bridge and culvert locations. Point features are also used to represent abstract points. For instance, point locations could represent city locations or place names.



In GIS, point data can be used to show the geographic location of cities. Map: Caitlin Dempsey using Natural Earth Data.

Line Data

Line (or arc) data is used to represent linear features. Common examples would be rivers, trails, and streets. Line features only have one dimension and therefore can only be used to measure length. Line features have a starting and ending point. Common examples would be road centerlines and hydrology. Symbology most commonly used to distinguish arc features from one another are line types (solid lines versus dashed lines) and combinations using colors and line thicknesses. In the example below roads are distinguished from the stream network by designating the roads as a solid black line and the hydrology a dashed blue line.

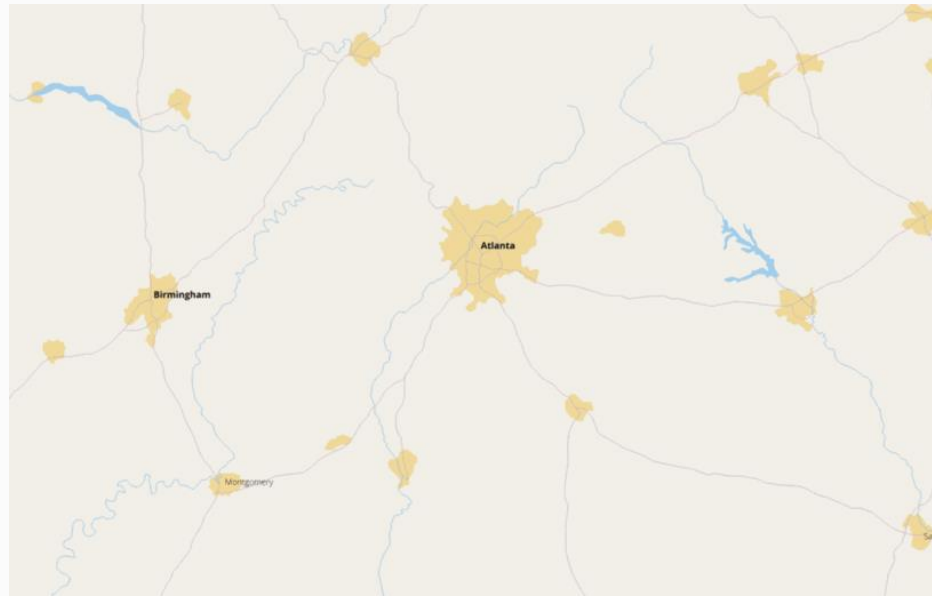


In this map, roads and waterways are shown as line data. Map using Natural Earth Data.

Polygon Data

Polygons are used to represent areas such as the boundary of a city (on a large scale map), lake, or forest. Polygon features are two dimensional and therefore can be used to measure the area and perimeter of a geographic feature.

Polygon features are most commonly distinguished using either a thematic mapping symbology (color schemes), patterns, or in the case of numeric gradation, a color gradation scheme could be used.



With maps presented at a larger scale, city locations are represented as a polygon to show the extent of each city. Map made with Natural Earth Data.

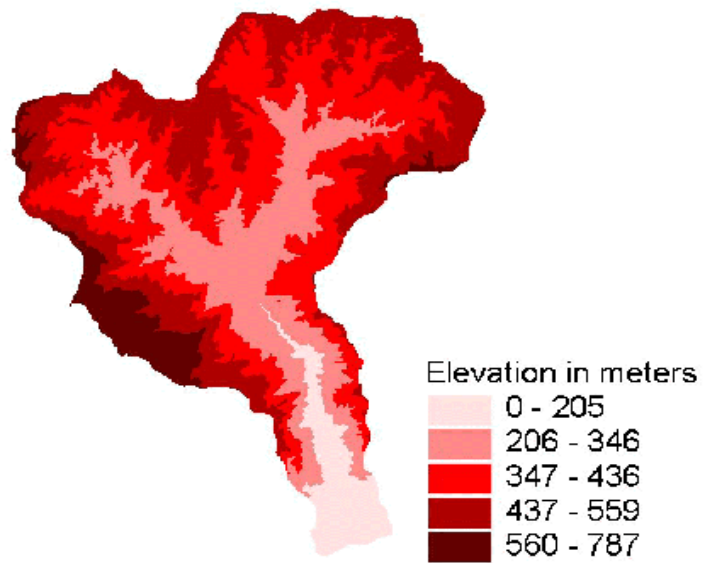
Both line and point feature data represent polygon data at a much smaller scale. They help reduce clutter by simplifying data locations.

As the features are zoomed in to, the point location of a school is more realistically represented by a series of building footprints showing the physical location of the campus.

Line features of a street centerline file only represent the physical location of the street. If a higher degree of spatial resolution is needed, a street curbwidth file would be used to show the width of the road as well as any features such as medians and right-of-ways (or sidewalks).

RASTER DATA

Raster data (also known as grid data) represents the fourth type of feature: surfaces. Raster data is cell-based and this data category also includes aerial and satellite imagery. There are two types of raster data: continuous and discrete. An example of discrete raster data is population density. Continuous data examples are temperature and elevation measurements. There are also three types of raster datasets: thematic data, spectral data, and pictures (imagery).



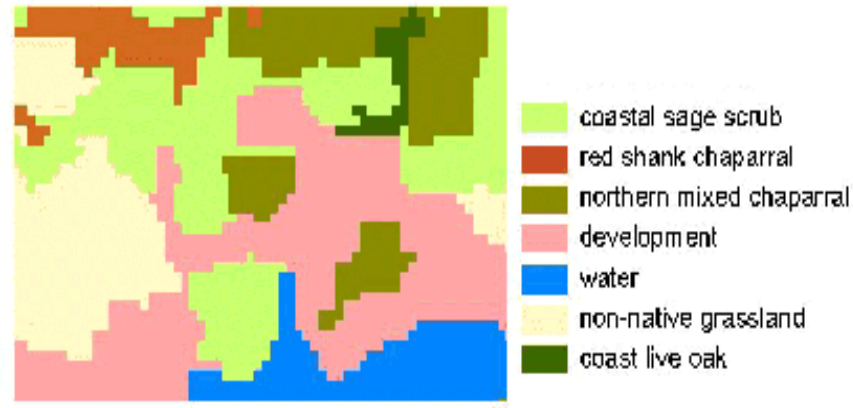
Digital Elevation Model (DEM) showing elevation.

This example of a thematic raster dataset is called a Digital Elevation Model (DEM). Each cell presents a 30m pixel size with an elevation value assigned to that cell. The area shown is the Topanga Watershed in California and gives the viewer and understand of the topography of the region.



This image shows a portion of Topanga, California taken from a USGS DOQ. Each cell contains one value representing the dominate value of that cell. Raster datasets are intrinsic to most spatial analysis.

Spatial hydrology modelling such as extracting watersheds and flow lines also uses a raster-based system. Spectral data presents aerial or satellite imagery which is then often used to derive vegetation geologic information by classifying the spectral signatures of each type of feature.

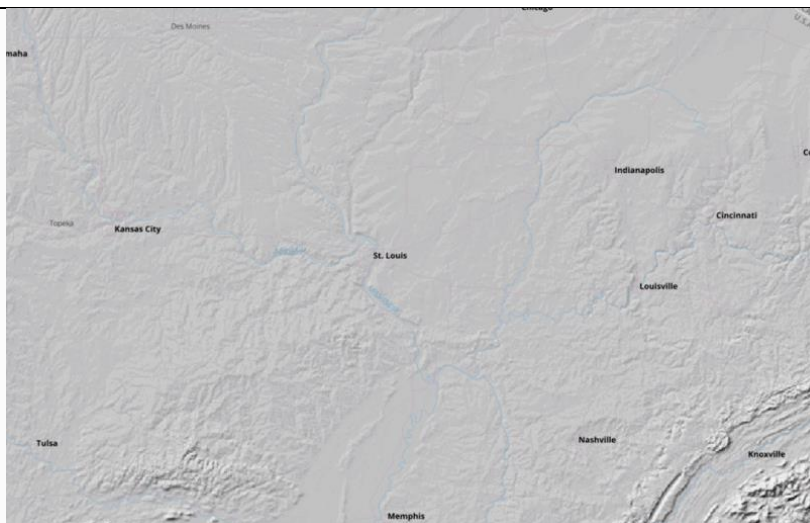


Raster data showing vegetation classification. The vegetation data was derived from NDVI classification of a satellite image.

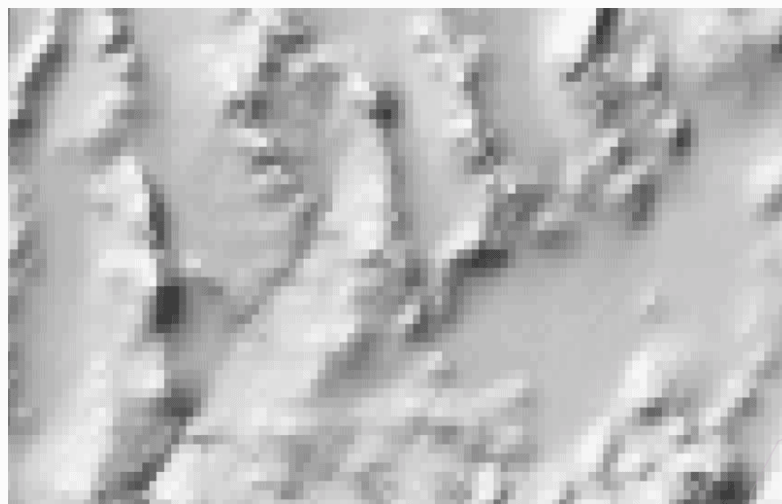
What results from the effect of converting spatial data location information into a cell based raster format is called stairstepping. The name derives from the image of exactly that, the square cells along the borders of different value types look like a staircase viewed from the side.

Unlike vector data, raster data is formed by each cell receiving the value of the feature that dominates the cell. The stairstepping look comes from the transition of the cells from one value to another. In the image above the dark green cell represents chamise vegetation. This means that the dominate feature in that cell area was chamise vegetation. Other features such as developed land, water or other vegetation types may be present on the ground in that area. As the feature in the cell becomes more dominantly urban, the cell is attributed the value for developed land, hence the pink shading.

Data analysis such as extracting slope and aspect from Digital Elevation Models occurs with raster datasets.



This 10 meters shaded relief was developed from downsampled SRTM Plus elevation data. Source: Natural Earth Data.



Zoom in closely to a raster dataset and you will see the individual cells. Image: Grayscale shaded relief of land areas derived from downsampled SRTM Plus elevation data. Source: Natural Earth Data.

Self-Evaluation

Name: _____ Date: _____

Program/Yr/Section: _____ Score: _____

Try to answer the questions below to test your knowledge of this lesson.

1. Why do we use GIS?
2. What is the main purpose of GIS?
3. What are the benefits of using GIS?

Review of Concepts	A geographic information system (GIS) is a spatial system that creates, manages, analyzes, and maps all types of data. It is a computerized instrument utilized for the purpose of cartography and analysis of spatially referenced phenomena and occurrences on the planet Earth. Spatial Information System (GIS) technology seamlessly combines conventional database operations, including query and statistical analysis, with the distinct advantages of visualization and spatial analysis provided by maps. A geographic information system (GIS) is basically a computerized information system like any other database, but with an important difference: all information in GIS must be linked to a geographic (spatial) reference (latitude/longitude, or other spatial coordinates). By querying the database associated to the layers, the GIS allows users to perform "why" and "what if" scenarios and creates a resulting map depicting a visual analysis. A working GIS integrates these five key components: hardware, software, data, people, and methods.
References	Majumder (ed). (2018). Application of Geographical Information Systems and Soft Computation Techniques in Water and Water Based Renewable Energy Problems. https://www.springer.com/gp/book/9789811062049 Favorskaya, et al. (2017). Handbook on Advances in Remote Sensing and Geographic Information Systems: Paradigms and Applications in Forest Landscape Modeling Tongco, A.F. & Tawas, S.K. (2018). Tutorials on Introductory Techniques Using Location-specific for Surigao del Sur Province, Philippines Intro to QGIS https://learn.canvas.net/courses/464 https://www.esri.com/en-us/what-is-gis/overview http://ecoursesonline.iasri.res.in/mod/page/view.php?id=2069 https://www.uou.ac.in/sites/default/files/slm/GIS-503.pdf